

# 1 Experiments 1-4

To try to understand the greater by-content differentiation of the ‘asking whether’ diagnostic, we focus on properties that set it apart from the other three diagnostics. Two differences are particularly salient: First, it is the only diagnostic where the target content itself occurs in a polar question, whereas in the other diagnostics, the target content appears in a declarative sentence. Second, it is the only diagnostic where participants are asked directly about the intentions of the speaker, while the other ones collect meta-linguistic judgments like naturalness ratings or continuation preferences. To distinguish between these possibilities, the next section compares the ‘asking whether’ diagnostic (Exp. 5) to a diagnostic that also embeds the target content in a polar question but elicits naturalness ratings on direct responses (Exp. 6).

## 1.1 Speech act, at-issueness, and speaker commitments

**JT: I’ve rewritten this section, see the next section, but I’ve also left some comments in this section**

The empirical generalization that diagnostics that present target contents in declarative sentences show reduced differentiation, whereas diagnostics using polar questions show greater differentiation raises the question of why the embedding context should have such a strong effect on diagnostic outcomes. We hypothesize that this pattern reflects a ceiling effect in assertion-based diagnostics, driven by how different speech acts relate the main-clause content to speaker commitment. We assume that the declaratives in these diagnostics are interpreted as assertions, which involve speaker commitment to the truth of the at-issue main-clause content; and the polar questions are interpreted as information-seeking questions, which do not involve speaker commitment. Because content that is less at-issue is more (likely to be) projective (Tonhauser et al. 2018; Degen & Tonhauser 2025), not-at-issue content is also associated with speaker commitments. **JT: This statement cannot be true for declaratives across the board. For instance, in “Cole is right that c” or “Cole knows that c”, the not-at-issue content is a speaker commitment because it is entailed, not because it is projective (as there is no entailment-canceling operator to project over).** As a result, the relationship to speaker commitments of not-at-issue content differs more sharply from the at-issue main clause content in polar questions than in assertions.

To illustrate this key contrast, consider a polar question like (1), where the at-issue content determines the set of alternatives under consideration (1a), but the speaker is not committed to any of these alternatives. In contrast, not-at-issue content, like the NRRC content in (1b), is more likely to project, **JT: than which other content?** that is, more likely to be treated as true throughout the context set, and therefore as part of speaker commitments (Abusch 2010; Simons et al. 2010; Abrusán 2011; Tonhauser et al. 2018; Degen & Tonhauser 2025).

- (1) *Is Greg, who bought a car, envied by his neighbor?*
- a. At-issue contribution: determining the question alternatives  
{Greg, who bought a car, is envied by his neighbor,  
Greg, who bought a car, is not envied by his neighbor}
  - b. Not-at-issue contribution: *Greg bought a car.*

As a result, the at-issue and not-at-issue content of questions differ sharply in their *epistemic status* relative to the speaker commitments: their not-at-issue content is associated with speaker commitment, while their at-issue content is not.

In assertions, this contrast is weakened. An assertion of a declarative sentence like (2) both proposes to add the content of the at-issue main clause (2a) to the common ground and commits

the speaker to its truth (Stalnaker 1978; Gunlogson 2001). In addition, the not-at-issue content (2b) is also treated as part of speaker commitments.

- (2) *Greg, who bought a car, is envied by his neighbor.*
- a. *Greg is envied by his neighbor.* (at-issue main clause content)
  - b. *Greg bought a car.* (not-at-issue content)

In (2), both the at-issue and not-at-issue content are associated with speaker commitment. The epistemic distinction between the two types of content is therefore reduced.

We propose that this difference in epistemic contrast could affect how participants identify at-issue content in experimental tasks. Diagnostics of at-issueness require participants to evaluate which content an utterance is primarily about. In assertions, the at-issue main clause content is the content to which the speaker is committed. Participants may therefore use speaker commitment as a cue for identifying the main point of the utterance. **JT: This is a HUGE hypothesis. And I'm not sure it stands: In sentences like 22, where the speaker is committed to both contents, Syrett/Koev found a clear at-issueness difference between the two contents.** However, not-at-issue content, such as appositive material, is also typically treated as part of speaker commitments, due to projection. As a result, both the at-issue main clause and not-at-issue content satisfy the same criterion: they are contents the speaker is committed to. We suggest that this similarity makes the two types of content harder to distinguish in assertion-based diagnostics. This would lead to not-at-issue content being rated more similarly to at-issue content. **JT: Again, this is not what Syrett/Koev found giving rise to overall higher ratings for not-at-issue content JT: I don't understand what "higher ratings" means here; note how I got rid of all of the "higher" and "lower" wordings in section 1 when describing the diagnostics** and reduced differentiation across contents, essentially creating a ceiling effect.

This reduction in contrast does not arise in polar questions: Although the not-at-issue content of questions is also associated with speaker commitment due to projection, their at-issue content is not. Therefore, speaker commitment is not a cue for identifying the main point of the utterance in questions. **JT: another HUGE hypothesis** and interpreters must rely on other aspects of the utterance instead. Since the epistemic contrast between the two types of content is preserved in questions, ratings for not-at-issue content are not affected by any similarity to the at-issue main clause content, allowing for greater differentiation overall differentiation. On this view, polar questions do not so much enhance differences as leave them visible, whereas assertions tend to mask them.

Importantly, this hypothesis does not depend on the target content being clearly not-at-issue, as in the case of appositive NRRCs. **JT: Syrett/Koev suggest that NRRCs aren't "clearly not-at-issue"** but it predicts that assertion-based diagnostics can obscure more subtle differences in at-issueness as well. For contents whose status is less clearly determined by conventional meaning, such as complements of verbs like *discover* or *reveal*, reduced epistemic contrast in assertions may compress differences between at-issue and not-at-issue content as well. In the context of a diagnostic, the assertion of (3) signals that the speaker is committed to the main clause at-issue content (3b), and we would be testing the at-issueness of the target content (3a).

- (3) *Helen confirmed that Tony had a drink last night.*
- a. *p : Tony had a drink last night* (target content)
  - b. *q : Helen confirmed p* (main-clause content)

Now, if (3a) (p) is interpreted as more at-issue, at-issueness ratings will be higher. If p is interpreted as less at-issue, at-issueness ratings should in principle be lower, but the content is also more likely to project, leading to speaker commitment **JT: so you're assuming that clause-embedding predicates are entailment-canceling operators from under which the CC projects? because**

otherwise, there is no operator in examples like 23 from under which the CC could be claimed to project. while I see how that works for “confirm” or “believe”, does it work for “discover”, “be right”, and “know”? (Tonhauser et al. 2018; Degen & Tonhauser 2025). The idea here is that ratings for less at-issue content are inflated. **JT: “inflated” means that the ratings suggest that the content is more at-issue than it actually is?, see comment on “higher” and “lower” above** because not-at-issue content is associated with speaker commitment and therefore receives higher ratings from being harder to distinguish from at-issue content.

The hypothesis laid out here can explain the main pattern observed in our experiments. Diagnostics that embed target contents in polar questions, such as the ‘asking whether’ diagnostic (Exps. 2, 5) and the direct-response diagnostic (Exp. 6), preserve the epistemic contrast between at-issue and not-at-issue content, and are therefore suitable for detecting underlying differences in at-issueness associated with different expressions, yielding high by-content differentiation. Diagnostics that embed target contents in assertions, on the other hand, such as the QUD diagnostic (Exp. 1) and the dissent-based diagnostics (Exps. 3–4), compress differences between at-issue and not-at-issue-content, and yield lower overall differentiation. **JT: Why did the contrast emerge in Syrett/Koev?**

To our knowledge, no previous work has proposed that the illocutionary force of the context in which target content is presented can itself be a major driver (or inhibitor) of differentiation. Here, we saw that polar questions more transparently reveal how different contents can be interpreted as at-issue or not-at-issue, and suggested that this may be because the two contents differ in their epistemic status. Assertions, in contrast, commit the speaker both to the at-issue assertion and the not-at-issue content, thereby obscuring the distinction. If this is on the right track, diagnostics that embed the target content in questions could be understood as more sensitive to fine-grained differences in at-issueness associated with different expressions, whereas assertions introduce a confounding factor: the speaker commits to both types of content, reducing differentiation in at-issueness judgments.

## Why do the different experiments/diagnostics give different results?

Possible explanations:

- i. For the CCs of the five clause-embedding predicates: Exp 2 shows great differentiation (know < confess, discover < confirm < be right; 9/10 cells), Exp 3 shows no differentiation (0/10 cells, and Exps 1 (6/10 cells, but be right wonkiness) and 4 (5/10) show much less differentiation
  - **Hypothesis:** The greater differentiation in Exp 2 is because of the embedding polar questions, not because of the response task.
  - We confirmed this by running Exps 5-6, where we found lots of differentiation among the CCs of the 20 clause-embedding predicates.
  - Why would polar question-embedding produce greater differentiation than declarative embedding in the five predicates featured in our experiments?
  - Observation: We picked the five predicates because they differed greatly in the asking-whether diagnostic. But the CCs of all five predicates received pretty high ratings in gradient and categorical inference experiments of Degen/Tonhauser/Language, so there’s a pretty strong inference that the CC follows from these utterances (not all entailed; confess < know < confirm < discover < be right); only one that is entailed is be right and know.
  - So the five predicates don’t differ a great deal in declaratives when it comes to speaker commitment (? or interpreter inference) to the CC

- So when a speaker in Exps 1, 3–4 asserts the declarative, the interpreter is likely to take both contents to follow. These three diagnostics may not reliably distinguish between at-issueness but are rather about availability for subsequent reference. And since all of the CCs are (almost) entailed, they are available for subsequent reference.
- Exp 2 is the only diagnostic where MC and CC are both embedded under entailment-canceling operator; so here we entailments are canceled and the diagnostic can pay more attention to at-issueness? the more projective the more not-at-issue
- [Scontras & Tonhauser 2025; 2026](#): q-know utterance has more utility when CC is assumed to be true than when BEL is assumed to be true.  $\implies$  explains low at-issueness rating for CC in Exp 2

## 1.2 Diagnostic differences and notions of at-issueness

Because the diagnostics tested in this paper are grounded in different assumptions about what it means for content to be at-issue, the observed differences between them bear on the broader question of whether diagnostics track a single underlying notion of at-issueness or operationalize distinct phenomena. This subsection discusses how the observed differences relate to patterns that would be expected based on three positions: the unified view, on which different diagnostics target a single unified phenomenon of at-issueness (e.g., [Tonhauser 2012](#); [AnderBois et al. 2015](#); [Faller 2019](#)), the dual-notion view, on which they target different underlying notions ([Koev 2018](#)); and the anaphoric-accessibility view, on which some diagnostics target at-issueness, while others diagnose the availability of the target content for propositional anaphora ([Snider 2017a; a; 2018](#); [Korotkova 2020](#)). **JT: Like I wrote above, I don't understand how the dual-notion and the anaphoric-accessibility views differ since both assume that the various diagnostics target different underlying notions.** These positions differ with respect to how different diagnostics relate to the two definitions of QUD at-issueness and proposal at-issueness, repeated here:

- ?? QUD at-issueness: (based on [Simons et al. 2010](#), p. 323)  
 A content  $m$  is at-issue in a context  $c$  iff
- $m$  is relevant<sup>1</sup> to the QUD in  $c$ , and
  - the speaker has a recognizable intention to address the QUD via  $m$ .
- ?? Proposal at-issueness: ([Koev 2013](#), pp. 50–51)  
 A proposition  $p$  is at-issue in a context  $c$  iff
- $p$  is a proposal in  $c$  and
  - $p$  has not been accepted or rejected in  $c$ .

The literature differs in how the theoretical assumptions behind these two definitions should be interpreted. And the different positions lead to different expectations about what kind of empirical divergence would count as evidence to support them. The unified view would be supported by convergent results across diagnostics, but it is also consistent with divergence, provided that any diagnostic differences can be explained by factors independent of at-issueness. In contrast, the dual-notion view predicts that at least some diagnostic differences reflect a split between the diagnostics assumed to align with the proposal-based definition (i.e., the dissent-based diagnostics from Exps. 3–4) and those more in line with the QUD-based definition (i.e., the QUD diagnostic from Exp. 1; the ‘asking whether’ diagnostic from Exps. 2 and 5; and the direct-response diagnostic from Exp. 6). Finally, the anaphoric-accessibility view predicts systematic differences between diagnostics that involve propositional anaphora, like the dissent-based diagnostics (Exps. 3–4) and the direct-response diagnostic (Exp. 6), and those that do not, like the QUD diagnostic (Exp. 1) and the ‘asking whether’ diagnostic (Exps. 2 and 5).

In the following, we discuss whether our results provide support for any of these views.

The clearest diagnostic difference we found—whether target contents are presented in polar questions or declaratives—may appear related to the distinction between QUD-based and proposal-based definitions of at-issueness, since both involve a contrast between questions and assertions. However, as discussed in § ??, the QUD-diagnostic (Exp. 1) assumes the QUD-based definition but presents the target content in a declarative sentence, distinguishing between the factors of question embedding and QUD-at-issueness. Contrary to what would be expected based on the dual-notion view, it patterned with the dissent-based diagnostics (Exps. 3 and 4) in showing low by-content differentiation. The observed pattern is also not expected based on the anaphoric-availability view: The direct-response diagnostic from Exp. 6 uses the anaphoric polarity particle *yes* to diagnose at-issueness, but it shows high by-content differentiation and does not pattern with the dissent-based diagnostics which also involve propositional anaphora. **JT: But the anaphoric-accessibility view could be correct and that would still be compatible with the pattern we found because there are so many other factors that could modulate whether the diagnostic produces differentiation. Given all these factors, I'm not sure whether two diagnostics showing the same pattern can be used to argue for anything, or two diagnostics showing a different pattern.** Therefore, the main contrast we found cuts across the groupings expected under views that assume distinct underlying phenomena. As argued in § (3), this contrast can be explained independently of differences in at-issueness. While all three positions are in principle compatible with some diagnostic differences that are independent of at-issueness, the dual-notion or anaphoric-availability views do lead to expectations about patterns of differences between diagnostics that are not supported here. However, the contrast could be interpreted as most consistent with a unified view, which more directly predicts that any diagnostic differences are due to independent factors.

While not all diagnostic differences we found are explained by the main contrast between question and declarative embeddings, the remaining differences also do not align with the patterns that would be expected based on the different theoretical views. Instead, they appear to reflect diagnostic-specific effects tied to particular contents and response tasks. For instance, the exceptional behavior of *be right* under the QUD diagnostic does not plausibly reflect a difference in at-issueness either. The low ratings more plausibly arise from a pragmatic incompatibility. Our hypothesis here was that *be right* requires a discourse structure that conflicts with the assumptions made by the diagnostic (see discussion in § ??). As a result, the QUD-diagnostic (Exp. 1) differs from all other diagnostics in its ranking of *be right*. Again, this result is in principle compatible with all three views, but it not lend support to the dual-notion or anaphoric-availability views. Instead, it highlights how diagnostic differences can arise from interactions with expression-specific contextual requirements, which, again, would be the only interpretation that is consistent with the unified view.

Further, some small differences between diagnostics can be attributed to response-task effects. This is evident in the subtle differences between Exps. 5 and 6 (see § ??), which differed only in response format, and in our failure to replicate Syrett & Koev's 2015 reported positional effect for appositive NRRCs (see § ??). While they found sentence-final appositives to be more at-issue than sentence-medial ones under a direct-dissent diagnostic, our dissent-based *yes*, *but* diagnostic in Exp. 4 showed a difference in the opposite direction, and Exps. 1–3 found no positional difference. Taken together, these results provide no evidence for a stable notion of at-issueness under which sentence-final appositives are systematically more at-issue than medial ones (or the other way round), and they highlight how even small task differences can affect outcomes.

In sum, none of the diagnostic differences we found do not pattern along the lines expected based on the views suggesting that different at-issueness diagnostics are sensitive to distinct underlying phenomena. While this does not provide conclusive evidence to rule out the possibility that some diagnostics are sensitive to distinct underlying notions, it does suggest that much of the observed variation arises from factors other than distinct notions of at-issueness, namely differences in illocutionary force, contextual felicity requirements, and response-task effects.

The anaphoric-accessibility view (Snider 2017b; a; 2018; Korotkova 2020) shares the assumption that differences between diagnostics arise from interactions with expression-specific contextual requirements rather than from differences in at-issueness itself. In particular, it highlights that the direct-dissent and ‘yes but’ diagnostics rely on propositional anaphora and therefore impose specific constraints on discourse structure, common ground, and anaphoric accessibility. These constraints can independently affect acceptability judgments. However, the direct-response diagnostic (Exp. 6) also involves propositional anaphora (i.e., the polarity particle *yes*) but does not pattern with the dissent-based diagnostics (Exps. 3–4), suggesting that differences between diagnostics cannot always be reduced to whether or not they involve propositional anaphora.

Consequently, our results do not decisively rule out any of the three views discussed here, they can be taken to tentatively suggest that the observed diagnostic differences are most in line with a unified view of at-issueness, where different diagnostics offer different, partially overlapping windows onto a single underlying phenomenon. A final piece of evidence that also tentatively points in favor of this view is that, the diagnostics also showed convergence in the overall rank ordering of contents. For example, *pretend* and *see* consistently ranked low across diagnostics, while *discover* and *know* consistently ranked high. For instance, among most pairs from Exps. 1–4, the Spearman rank correlation was at least moderate (when excluding *be right*, see ??), suggesting some convergence among diagnostics, and that the diagnostics, as implemented there, are at least partially sensitive to a shared underlying property.

## 2 Conclusion

This paper presented a systematic experimental comparison of at-issueness diagnostics, motivated by two overarching questions: whether different diagnostics yield consistent results when applied to the same contents, and whether any differences between them support the view that they are sensitive to distinct underlying phenomena.

Across six experiments, we tested five diagnostics: the QUD diagnostic, the ‘asking whether’ diagnostic, the direct-dissent diagnostic, the ‘yes, but’ diagnostic, and the direct-response diagnostic. Our results provide experimental confirmation for claims from the theoretical literature that at-issueness diagnostics differ empirically (Snider 2017b; a; 2018; Koev 2018; Faller 2019; Korotkova 2020). Since they did not yield consistent results, they cannot be treated as interchangeable.

We also identified a clear source of variation: diagnostics that present target contents in polar questions showed greater by-content differentiation than those presenting the same contents in declaratives. We argued that this pattern reflects the role of illocutionary force and speaker commitments, and that these factors should be considered when interpreting diagnostic outcomes.

Our findings do not support a view that different diagnostics are sensitive to distinct underlying phenomena. Although the diagnostics diverge empirically, the observed differences did not align with existing distinguishing QUD-based at-issueness, proposal-based at-issueness, or anaphoric availability. Instead, they can be attributed to factors independent of the proposed theoretical distinctions, including speaker commitments, illocutionary force, contextual felicity constraints, and the given alternatives in forced-choice tasks. The results are therefore more consistent with a unified view, on which diagnostics provide partially overlapping perspectives on a single discourse phenomenon that interacts with multiple factors in complex and often subtle ways.

However, our results do not conclusively rule out the possibility that some diagnostics may be sensitive to distinct underlying notions of at-issueness. Determining what different diagnostics can reveal about whether they are sensitive to a shared underlying notion therefore remains an important question for future work.

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